

SCALING ATTENTION-BASED MODELS THROUGH DISTRIBUTION-AWARE GROUP QUANTISATION

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ABSTRACT

We reproduce DGQ Ryu et al. (2025), a training-free post-training quantisation method for text-to-image diffusion models that highlights two challenges: maintaining outlier information when quantising activations, and prompt-dependent cross-attention score distributions that static scales cannot cover. We implement and extend DGQ’s group outlier quantisation to three additional architectures: Flux2 Klein (a Diffusion Transformer), Gemma 3 4B (a Vision-Language Model), and YOLOv8n (a CNN object detection model). Across all extensions, our results suggest that DGQ’s core distributional insight generalises beyond UNet-based diffusion models to a broad class of pre-trained deep networks. All code to reproduce our experiments and figures are found in the GitHub repository: <https://github.com/bboyapati/COMP6258-Coursework>

1 INTRODUCTION

Post-training quantisation (PTQ) compresses models into low-bit integer formats without retraining. For diffusion models, prior methods such as Q-Diffusion Li et al. (2023) and TFMQ-DM Huang et al. (2024) fail below 8-bit precision. DGQ identifies the root cause by analysing the distributions of activations; UNet activations contain structured outliers that contain critical information that is lost when quantising broadly.

We aim to reproduce DGQ’s methodology and test whether the distributional insight generalises beyond UNet-based diffusion models to various models.

2 METHOD AND DISTRIBUTIONAL ANALYSIS

DGQ identifies a challenge through distributional analysis of *StableDiffusion v1.4*:

C1: Activation outliers Outliers are defined as small fractions of activations with either very high or very low values; the paper argues that setting a single outlier activation to zero per layer produces images with drastically different shapes and lower quality, whereas dropping a random activation has minimal impact.

The paper further shows that outliers concentrate on specific pixels (*pixel-wise*) or specific channels (*channel-wise*), persisting across seeds and prompts.

DGQ. To address C1, outlier channels and pixels are assigned independent group-wise scale parameters with non-outlier activations sharing a coarser grouping.

3 EXTENSION TO DIFFUSION-TRANSFORMER (DiT) MODELS

Diffusion Transformers (DiT) replace the UNet denoising backbone entirely with a Vision Transformer operating in latent space. Unlike SD v1.4, there is no explicit cross-attention layer separating text from image; conditioning is handled through joint attention over concatenated token sequences.

Setup We extend DGQ to Klein 4B, a DiT-based diffusion model due to our hardware constraints (Iridis L4 GPUs). The model operates using the bf16 datatype, which is a specific format of 16-bit floating point. The implementation was built on top of the existing code of DGQ, which required implementation of wrapper classes to wrap the main Flux2 attention and transformer block from the updated diffusers class.

DGQ has predefined wrappers for Linear and Conv layers under the QuantLayer class, which wraps the weights, inputs and the activations under UniformAffineQuantizers. Therefore, it does fake quantisation which stores the values in their original datatype despite the quantisation being done to a lower bit count. This allows for the computation to continue without converting datatypes as the latent variable is decoded to the image space via the VAE.



(a) Flux 2

(b) Flux 2 w4a6

Figure 1: Example Images between Flux2 vs Quantised Flux2 with w4a6

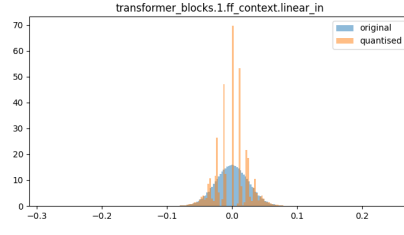


Figure 2: Comparison of weights between original model and quantised model

Results Figure 1 shows some examples of outputs between the original model and the model quantised with 4 bit weights and 6 bit activations with the same seed. Between the images of the virus monster, the original model has a wider variety of colours as seen in the background and the monster’s skin compared to the quantised model. The pose could also be seen as more dynamic with the one leg kneeling down and the angle of the body as opposed to the more neutral pose of the quantised model. Figure 2 shows an example of the difference between weights at some layers of the diffusion transformer. The quantisation can be seen as the weights are more discretised compared to the more continuous distribution of the original model.

4 VISION-LANGUAGE MODELS

Setup We focus solely on the Language Model. As DGQ is designed for text-to-image models, various modifications have had to be made to quantise LLMs. These experiments were done using Google’s Gemma 3 4B parameter model due to VRAM constraints. The MMLU-Pro dataset was the primary dataset used for calibration and evaluation Wang et al. (2024).

4.1 OUTLIER-PRESERVING GROUP QUANTISATION

The changes made for this method to work on LLMs were to drop the idea of pixel-level outliers and variance of activations across diffusion timesteps due to the difference in architecture and instead, focus on channel outliers as LLMs are known for having large, persistent outliers in specific channels in literature Dettmers et al. (2022). Instead of using K-means clustering, after running a calibration dataset through the model to identify outlier channels, a ”outlier fraction” was set as a parameter for the percentage of channels to mark as outliers. As these outliers are persistent even when seed and input is different, it was cached in a JSON file. It was noted that a high enough ”outlier fraction” has to be used to map all outlier as these outliers are not perfectly persistent. The result of this is marginally faster generation speeds

4.2 ATTENTION-AWARE QUANTISATION

This translates directly to LLMs. LLMs have the same issue and is more commonly known as the Attention Sink Phenomenon Gu et al. (2025). The main difference is to apply the quantisation to the

KV-cache rather than the attention matrix itself to prevent breaking flash attention and consequently, possible crippling the generation speed. The KV-cache of the “<bos>” token must be kept at a higher precision while the rest can be quantised to a lower precision. As a result of this quantisation method, quantising the rest of the KV-Cache to 4 bit reduced performance on the MMLU-Pro dataset by a great margin, however quantising at 8 bit instead marginally increased performance for higher precision quantisations (W8A8, W8A6 and W4A8).

KV-Cache Quant (Bits)	W8A8	W8A6	W4A8	W4A6
16	22.52%	21.69%	18.18%	15.44%
8	22.52%	22.32%	19.18%	14.86%
4	22.52%	18.30%	14.18%	6.89%

Table 1: KV-Cache, Weights and Activations were quantised and MMLU-Pro was used to benchmark the model. Table shows percentage of the questions answered correctly by the quantised model.

5 SCALING GROUP QUANTISATION TO YOLOV8

To evaluate whether DGQ’s outlier-preserving group quantisation generalises beyond text-to-image diffusion models, we implemented our solution of DGQ to a YOLOv8n Jocher et al. (2023) model.

Setup We registered forward hooks on all 72 convolutional layers except the final one, and ran 64 MS-COCO val2017 images through the model to capture real activation distributions to match the paper’s 64 sample calibration procedure. For each layer, we create a wrapper, that quantises its weights on initialisation, and quantises the activations during the forward method. Activations are put into 2 groups by K Means, matching the implementation by DGQ, meaning that outliers have unique quantisation parameters. We store these groups to expedite model quantisation.

Results Preliminary results show that the quantised models can perform as well as a full-precision (32 bits) model. We see that quantising the weights and activations at 4 bits or below incurs serious performance costs, leading to a more balanced quantisation scheme, such as 8 bits for weights and activations.

We find that not quantised models consistently perform better, but this is most likely because they are highly optimised for 32 bit or 64 bit hardware, rather than the lower bits used on edge devices. Further, quantising a model increases its variance Nagel et al. (2021), meaning results can be more unstable, as we can see in the example predictions from YOLOv8 detection architecture.



Figure 3: YOLOv8 predictions across varying levels of quantisation predicting the object in a complex scene containing: 13 people, 1 bus 1 car and 1 handbag. The figure on the left is not quantised: It identified 10 people, 2 cars and 1 bus. The image in the middle has 6 bit weight quantisation and 8 bit activations: Identified 13 people, 1 car and 1 truck. The final picture has 8 bit weights and 8 bit activations: Identified 10 people, 2 cars, 1 bus and a handbag.

Interpretation These results suggest that activation outliers are a general property of deep networks rather than a phenomenon strictly specific to diffusion models Bondarenko et al. (2021). Channel-wise quantisation granularity, which underlies DGQ’s grouping, is already established as the standard approach for CNN quantisation Nagel et al. (2021), suggesting the method’s assumptions about inter-channel variance generally hold for convolutional models.

6 CONCLUSION AND FUTURE WORK

We successfully reproduce outlier-preserving group quantisation and confirm that activation outliers are both locally structured and causally important for image quality, as the paper claims. For the DiT model, DGQ maintained quality in the images, but there could be a reduction in the level of detail in the generated images. DGQ generalised well for the YOLO model.

With more compute resources, we could use the original models instead of distilled models. For example, the original Flux2 model is 32B parameters and would take ≥ 64 GB of VRAM. Also a comparison of FID or CLIPScore metrics between different bit sizes and the original model would require a large amount of time to generate a sufficient sample set for a reliable measure. By converting the datatype of the transformer’s processes (activations and weights) to the lower bit format instead of just representing these lower bit values in the original data type would allow for the model size to decrease. Implementing this would allow for realised reduction of VRAM usage of the models.

VLMs usually have a vision encoder head to encode images to be passed into the LLM. VLMs could benefit from this encoder head being quantised by reducing latency and/or reducing time to process token. In our YOLO extension work, we made assumptions concerning the calculation of optimal dimension. We adapted the quantisation scale and zero-point equations as the provided formula were not reproducible in our case.

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